Toolchain Smoke Test for a Stochastic-Computing TPU Coupled to a RISC-V CPU

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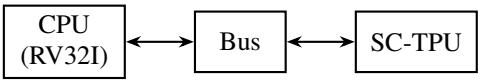


Fig. 1. Block diagram: CPU connected to the TPU through a bus.

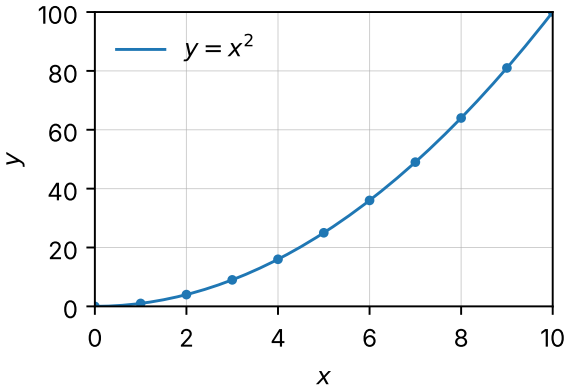


Fig. 2. Synthetic smoke-test plot of $y = x^2$ for $x \in [0, 10]$.

***Abstract*—This one-page document exercises the documentation toolchain of the project: the IEEEtran conference class, an equation, a TikZ block diagram, a vector figure produced by matplotlib, and BibTeX citations. It reports no experimental or simulation results.**

I. Introduction

Stochastic computing (SC) represents a value $p \in [0, 1]$ as the probability that a bit of a random stream is one [1], [2]. For two independent streams A and B, a single AND gate computes the product

$$P(A \wedge B) = P(A)\,P(B), \tag{1}$$

which is the basic operation of an SC multiply-accumulate array. Equation (1) is the textbook identity for independent events; it is not a project result. Fig. 1 shows the intended top-level organisation, in which a CPU drives the tensor processing unit (TPU) over a shared bus, and Fig. 2 is a synthetic test plot of $y = x^2$ (CALCULATION; generator: `reports/figures/smoke_plot.py`).

References

[1] B. R. Gaines, "Stochastic computing systems," in *Advances in Information Systems Science*, J. T. Tou, Ed. New York, NY, USA: Plenum Press, 1969, vol. 2.

[2] A. Alaghi and J. P. Hayes, "Survey of stochastic computing," *ACM Transactions on Embedded Computing Systems*, vol. 12, no. 2s, 2013.